Law of Sines and Law of Cosines (TES)

Math 172 lecture notes · Section 4.3.2

Learning objective: use the Law of Sines and/or the Law of Cosines to solve non-right triangles.

Try each part before you open the answer. Your thinking is what makes it stick.

Problem from your notes *Text from the handout* *Extra note (not in your handout)*

Overview

FROM THE HANDOUT

We learned how to solve right triangles (and special right triangles) using trigonometric ratios. In this section, we will learn how to solve oblique (non-right) triangles.

Rounding

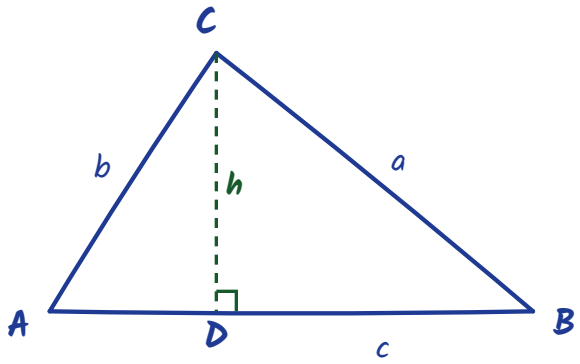
✎ EXTRA NOTE: NOT IN YOUR HANDOUT

Decimals shown in the middle of a problem are shortened so they're easy to read. Your calculator should keep the **full value** until the final answer. Rounding too early can change the answer.

Activity 1

Activity 1 · from your notes

Suppose you are given the following triangle, where h is the altitude of triangle ABC. By drawing the altitude, we've now created two right triangles. If we label the point where the altitude meets side c as point D, then we can call the triangles ACD and BCD.



(a) Find $\sin A$ and $\sin B$.

> **Hint:** Look at one right triangle at a time. Which side is opposite the angle? Which is the hypotenuse?

Answer (a)

In right triangle ACD, h is opposite A and b is the hypotenuse. In right triangle BCD, h is opposite B and a is the hypotenuse.

$$\sin A = \frac{h}{b}, \quad \sin B = \frac{h}{a}$$

(b) Take your answers and solve each equation for h (the altitude). What does h equal in each case?

Answer (b)

Multiply both sides by the denominator.

$$h = b \sin A \quad \text{and} \quad h = a \sin B$$

(c) Notice that we now have two ways to express h. Set the two expressions equal to one another and rearrange so that a and $\sin A$ are on the same side.

Answer (c)

$b \sin A = a \sin B$. Divide both sides by $\sin A \cdot \sin B$:

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

That's the Law of Sines!

Law of Sines

FROM THE HANDOUT

Theorem. The Law of Sines says the ratio of each side of a triangle to the sine of its opposite angle is the same for all three sides:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \quad \text{or} \quad \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

When using the Law of Sines, you will only need to set one fraction equal to one other fraction. Note that these are proportions and so the reciprocals also

hold true.

Note

FROM THE HANDOUT

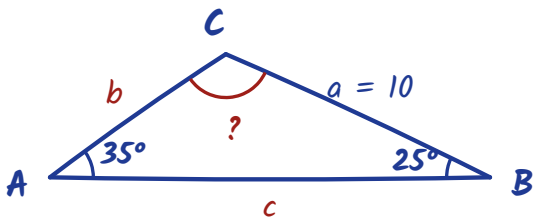
The Law of Sines is helpful when we have

- two angles and one of their corresponding sides, or
- two sides and one of their corresponding angles.

Activity 2

Activity 2 · from your notes

Suppose you are given triangle ABC where A = 35°, B = 25°, and a = 10 as shown in the figure.



(a) What parts of the triangle are missing?

Answer (a)

C, b, and c

(b) What is the value of C?

Hint: All triangle angles add up to 180°.

Answer (b)

$C = 180° - 35° - 25°$

C = 120°

(c) Use the Law of Sines to set up a proportion to find c. What does that proportion look like?

Answer (c)

Pair the side you know with its angle (a and A), and the side you want with its angle (c and C):

$$\frac{10}{\sin 35^\circ} = \frac{c}{\sin 120^\circ}$$

(d) Solve for c. What is the value of c to the nearest tenth?

Answer (d)

$$c = \frac{10 \sin 120^\circ}{\sin 35^\circ} \approx 15.098\ldots$$

$$c \approx 15.1$$

(e) We now have side b left to find. Which method would be the best method to find b?

Answer (e)

The Law of Sines again. This is not a right triangle, so the Pythagorean Theorem doesn't apply. Pair b with B, and use a known pair: a with A (the original, unrounded data), or c with C.

Law of Sines

(f) Use that method to find b. What is the value of b to the nearest tenth?

Answer (f)

$$\frac{10}{\sin 35^\circ} = \frac{b}{\sin 25^\circ} \rightarrow b = \frac{10 \sin 25^\circ}{\sin 35^\circ} \approx 7.368\ldots$$

$$b \approx 7.4$$

Note

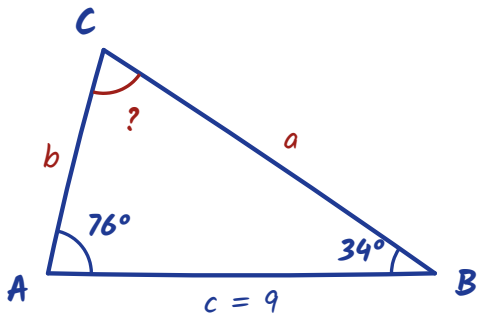
FROM THE HANDOUT

The triangle in the previous example is known as an **AAS** (Angle-Angle-Side) triangle, which means we know two angles and one side (which is NOT between the angles) in that order. This is just one type of triangle you might see when solving oblique triangles. The next activity will show another type of triangle you might encounter.

Activity 3

Activity 3 · from your notes

Suppose you are given triangle ABC where A = 76°, B = 34°, and c = 9 as shown in the figure.



(a) What parts of the triangle are missing?

Answer (a)

C, a, and b

(b) What is the value of C?

Answer (b)

$C = 180° - 76° - 34°$

C = 70°

(c) Use the Law of Sines to set up a proportion to find a. What is the value of a to the nearest tenth?

> **Hint:** You need a complete pair (a side and its opposite angle). Which pair can you complete now?

Answer (c)

Now that we know C, the pair c and C is complete:

$$\frac{a}{\sin 76°} = \frac{9}{\sin 70°} \rightarrow a = \frac{9 \sin 76°}{\sin 70°} \approx 9.293...$$

$$a \approx 9.3$$

Note

FROM THE HANDOUT

The triangle in the previous example is known as an **ASA** (Angle-Side-Angle) triangle, which means we know two angles and a side BETWEEN the angles.

Activity 4

Activity 4 · from your notes

Suppose you are given triangle ABC, where $A = 30°$, $a = 7$, and $b = 16$.

(a) What parts of the triangle are missing?

Answer (a)

$$B, C, \text{ and } c$$

(b) Use the Law of Sines to set up a proportion to find $\sin B$. What is the value of $\sin B$?

Answer (b)

$$\frac{\sin B}{16} = \frac{\sin 30°}{7} \rightarrow \sin B = \frac{16 \sin 30°}{7} = \frac{8}{7}$$

$$\sin B \approx 1.14$$

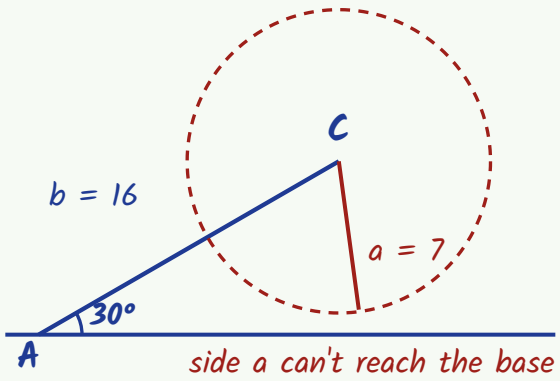
(c) Use part (b) to determine the value of B.

Answer (c)

$\sin^{-1}\left(\frac{8}{7}\right)$ gives an **error**: sine can never be bigger than 1.

No such angle B, so no triangle exists

The picture shows why: side a = 7 is too short to reach the base.



Note

FROM THE HANDOUT

In the previous example, we can see that no triangle can be created because sin B was equal to a value greater than 1. This is because the largest value that the sine of an angle can have is 1.

Activity 5

Activity 5 · from your notes

Suppose you are given triangle ABC, where A = 30°, a = 10, and b = 16.

(a) Draw triangle ABC and label what is known. What parts of the triangle are missing?

Answer (a)

B, C, and c

(b) Use the Law of Sines to determine the value of B (to the nearest degree).

Answer (b)

$$\sin B = \frac{16 \sin 30^\circ}{10} = 0.8 = \frac{4}{5} \rightarrow B = \sin^{-1}(0.8) \approx 53.13^\circ$$

$B \approx 53^\circ$

(c) Continue solving triangle ABC. What are the missing values (to the nearest degree/whole number)?

Answer (c)

$$C = 180^\circ - 30^\circ - 53.13...^\circ \approx 96.87^\circ. \quad c = \frac{10 \sin 96.87^\circ}{\sin 30^\circ} \approx 19.86$$

$C \approx 97^\circ,\ c \approx 20$

(d) In part (b) we solved $\sin B = \frac{4}{5}$ and the calculator gave $B \approx 53^\circ$. There is another angle between 0° and 180° whose sine is $\frac{4}{5}$. What could B also be?

Hint: Think about the unit circle: where else is the y-coordinate the same?

Answer (d)

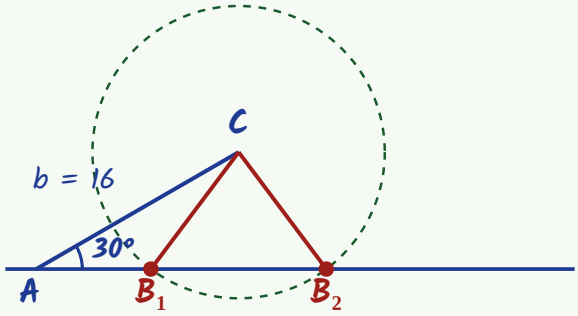
Sine is positive in Quadrant II too. The other angle is $180^\circ - 53.13^\circ = 126.87^\circ$.

$B \approx 127^\circ$

(e) Why do you think there are two values of B for this triangle?

Answer (e)

Sine is positive in Quadrants I and II, so two angles between 0° and 180° share the same sine. (For example, sin 30° = sin 150° = ½.) Geometrically, side a can swing to two different spots on the base.



(f) Continue solving this other triangle ABC. What are the missing values?

Answer (f)

$C = 180° - 30° - 126.87...° \approx 23.13°$. $c = \dfrac{10 \sin 23.13°}{\sin 30°} \approx 7.86$

$C \approx 23°$, $c \approx 8$

Note

FROM THE HANDOUT

So far we have seen that when given two sides and an angle (also known as an **SSA** (Side-Side-Angle) triangle), we can have no solution (i.e., no triangle can be created) or two solutions (i.e., there are two possible triangles). There is still one more case we need to explore.

Activity 6

Activity 6 · from your notes

Suppose you are given triangle ABC, where A = 30°, a = 20, and b = 16.

(a) Draw triangle ABC and label what is known. What parts of the triangle are missing?

Answer (a)

B, C, and c

(b) Use the Law of Sines to determine the value of B (to the nearest degree).

Answer (b)

$\sin B = \frac{16 \sin 30°}{20} = 0.4 \rightarrow B \approx 23.58°$, or $B \approx 180° - 23.58° = 156.42°$

Two candidates: $B \approx 24°$ or $B \approx 156°$

These are candidates. Part (d) tests whether the larger one can really fit in a triangle.

(c) Choose the smaller value of B and solve triangle ABC.

Answer (c)

$C = 180° - 30° - 23.58...° \approx 126.42°$. $c = \frac{20 \sin 126.42°}{\sin 30°} \approx 32.19$

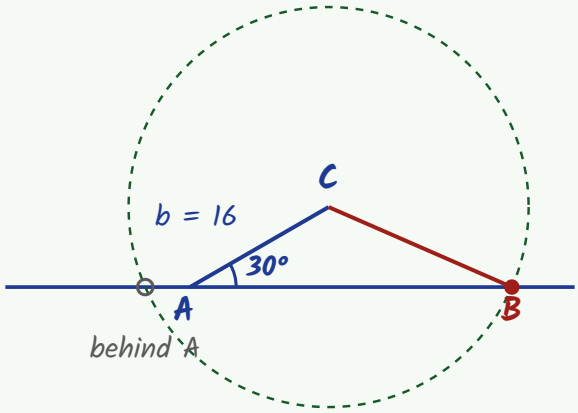
$C \approx 126°$, $c \approx 32$

(d) Choose the larger value of B and solve triangle ABC.

Answer (d)

$30° + 156.42° = 186.42°$, which is already more than $180°$. There's no room left for C.

No second triangle: exactly one triangle exists



Observation

FROM THE HANDOUT

In the previous three activities, we saw that when we use the Law of Sines to find an angle, an ambiguity can arise due to the sine function being positive in Quadrant I and Quadrant II. In other words, if two sides and the non-included angle are given (SSA), three situations may occur. The **Ambiguous Case** of the Law of Sines states that when using the Law of Sines to find a missing angle, there may be two possible solutions.

- NO triangle exists (no solution)
- TWO different triangles exist (2 solutions)
- Exactly ONE triangle exists (1 solution)

Your handout says "missing side length" here. It's the missing angle that can have two values.

Activity 7

Activity 7 · from your notes

State the number of possible triangles that can be formed with the given measurements. Then, solve each triangle. Round your answers to the nearest tenth.

(a) $C = 145°$, $b = 7$, $c = 33$

> **Hint:** Find the angle first. Then test the Quadrant II option: does it leave room for a third angle?

Answer (a)

$\sin B = \frac{7 \sin 145°}{33} \approx 0.1217 \rightarrow B \approx 7.0°$. The other option, $180° - 7.0° = 173.0°$, won't fit with $C = 145°$.

$A = 180° - 145° - 6.99° \approx 28.0°$. $a = \frac{33 \sin 28.01°}{\sin 145°} \approx 27.0$

One triangle: $B \approx 7.0°$, $A \approx 28.0°$, $a \approx 27.0$

(b) $B = 84°$, $b = 18$, $c = 9$

Answer (b)

$\sin C = \frac{9 \sin 84°}{18} \approx 0.4973 \rightarrow C \approx 29.8°$. The other option, $150.2°$, plus $84°$ is over $180°$.

$A = 180° - 84° - 29.82° \approx 66.2°$. $a = \frac{18 \sin 66.18°}{\sin 84°} \approx 16.6$

One triangle: $C \approx 29.8°$, $A \approx 66.2°$, $a \approx 16.6$

Activity 8

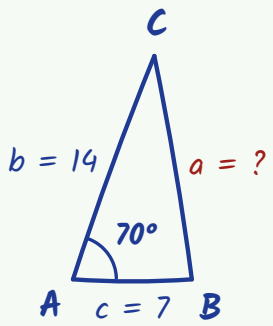
Activity 8 · from your notes

Suppose you are given triangle ABC, where $A = 70°$, $b = 14$, and $c = 7$.

(a) Draw triangle ABC and label what is known. What parts of the triangle are missing?

Answer (a)

B, C, and a



(b) Use the Law of Sines to solve triangle ABC.

Hint: List the pairs (a, A), (b, B), (c, C). Is any pair complete?

Answer (b)

We can't! Every Law of Sines proportion needs a complete pair: a side and its opposite angle. We know A, but not a. We know b and c, but not B or C.

The Law of Sines can't be used directly to find the first missing part

Note

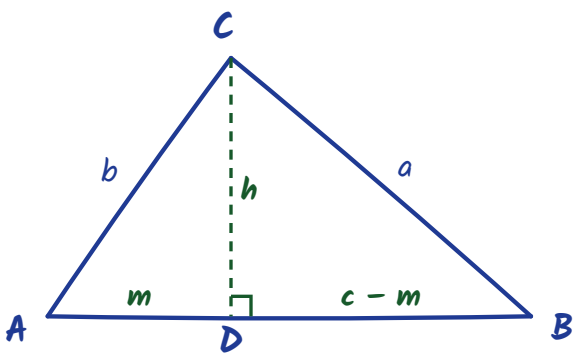
FROM THE HANDOUT

Notice in the previous example, we do not currently have enough information to be able to solve triangle ABC since the Law of Sines cannot be used. In the next activity, we will explore another method that can be used to solve oblique triangles.

Activity 9

Activity 9 · from your notes

Suppose you are given the following triangle, where h is the altitude of triangle ABC. The segment AD is labeled m, so DB = c − m.



(a/b) Using right triangle trigonometry, find sin A and cos A.

Answer (a/b)

$$\sin A = \frac{h}{b}, \quad \cos A = \frac{m}{b}$$

Your handout's description calls segment AD "d", but the parts below use m. We use m throughout.

(c) Take your equation for sin A and solve for h.

Answer (c)

$$h = b \sin A$$

(d) Take your equation for cos A and solve for m.

Answer (d)

$$m = b \cos A$$

(e) The Pythagorean Theorem for right triangle BCD is $h^2 + (c - m)^2 = a^2$. Substitute your expressions for h and m into that equation.

Answer (e)

$$(b \sin A)^2 + (c - b \cos A)^2 = a^2$$

(f) Simplify your equation from part (e).

> **Hint:** Don't forget that $(c - b\cos A)^2 = (c - b\cos A)(c - b\cos A)$. Then look for $\sin^2 A + \cos^2 A$.

Answer (f)

$b^2\sin^2 A + c^2 - 2bc\cos A + b^2\cos^2 A = a^2$

Group: $b^2(\sin^2 A + \cos^2 A) + c^2 - 2bc\cos A = a^2$

Pythagorean identity: $\sin^2 A + \cos^2 A = 1$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

Law of Cosines

FROM THE HANDOUT

Theorem. The Law of Cosines relates the lengths of the sides of a triangle to the cosine of one of its angles. In other words, the square of any one side of a triangle is equal to the sum of the squares of the other two sides, minus twice the product of those two sides and the cosine of the angle between them.

Let a, b, and c be the lengths of the three sides of a triangle and A, B, and C be the three angles. Then:

$$a^2 = b^2 + c^2 - 2bc\cos A$$
$$b^2 = a^2 + c^2 - 2ac\cos B$$
$$c^2 = a^2 + b^2 - 2ab\cos C$$

Activity 10

Activity 10 · from your notes

Let's revisit the previous example. Recall triangle ABC, where $A = 70°$, $b = 14$, and $c = 7$.

(a) Draw triangle ABC and label what is known. What parts of the triangle are missing?

Answer (a)

B, C, and a

This is an **SAS** triangle: the known angle is between the two known sides.

(b) Use the Law of Cosines to find a (to the nearest tenth).

Answer (b)

$a^2 = 14^2 + 7^2 - 2(14)(7)\cos 70° \approx 245 - 67.04 = 177.96$

$a \approx \sqrt{177.96} \approx 13.340\ldots$

$$a \approx 13.3$$

(c) Now that we have another side, what would you use to find angle B?

Answer (c)

Either law works now that the pair (a, A) is complete. The Law of Cosines is the safer choice: it gives the right angle even if B is obtuse, while the Law of Sines can't tell the two possible angles apart.

Law of Sines or Law of Cosines (Cosines is safer)

(d) Find the measure of B and C (to the nearest degree).

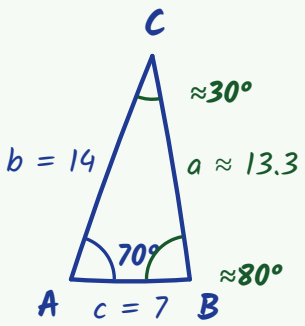
Answer (d)

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{177.96 + 49 - 196}{2(13.340)(7)} \approx 0.1658 \rightarrow B \approx 80.46°$$

$$C = 180° - 70° - 80.46° \approx 29.54°$$

B ≈ 80°, C ≈ 30°

> ⚠ **Watch out:** Use the unrounded a ≈ 13.340. If you round to 13.3 first, B comes out ≈ 81° and C ≈ 29°. That's how the textbook's answer key gets those values.



Activity II

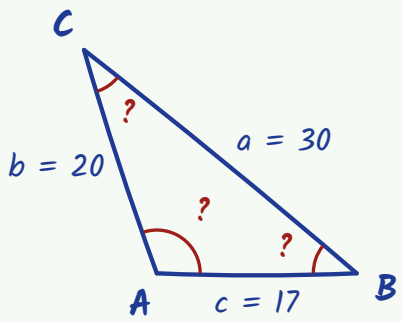
Activity II · from your notes

Suppose you are given triangle ABC, where a = 30, b = 20, and c = 17.

(a) Draw triangle ABC and label what is known. What parts of the triangle are missing?

Answer (a)

A, B, and C (all three angles)



(b) Which law should you start with to solve triangle ABC?

Answer (b)

No angle is known, so there's no complete pair for the Law of Sines.

Law of Cosines

(c) What is the value of A (to the nearest degree)?

Answer (c)

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{400 + 289 - 900}{2(20)(17)} = \frac{-211}{680} \approx -0.3103$$

$A \approx 108.08°$. A negative cosine means A is obtuse.

$A \approx 108°$

(d) Now that you have found angle A, which law would be the best to use to find angle B?

Answer (d)

The Law of Sines is quickest now that (a, A) is complete. It's safe here because B is across from a shorter side than a, so B must be acute. (The Law of Cosines works too.)

Law of Sines

If you keep going: $B \approx 39°$ and $C \approx 33°$.

Note

FROM THE HANDOUT

Notice in the previous example, you were given a triangle with three known sides. This type of triangle is known as a side-side-side (**SSS**) triangle.

Which law?

✎ EXTRA NOTE: NOT IN YOUR HANDOUT

Do I have a complete pair (a side and its opposite angle)?

- **Yes** (AAS, ASA, SSA): Law of Sines. With SSA, check for the ambiguous case.
- **No** (SAS, SSS): Law of Cosines first, then you'll have a pair.

Activity 12

Activity 12 · from your notes

State whether the Law of Sines or the Law of Cosines is the best choice to use to determine the indicated angle/side.

(a) Given $B = 112°$, $a = 12$, and $b = 25$, find A.

> **Hint:** It might help to draw a picture!

Answer (a)

(b, B) is a complete pair, and we know a.

Law of Sines

(b) Given $B = 87°$, $a = 15$, and $c = 16$, find b.

Answer (b)

B is between the two known sides (SAS). There's no complete pair.

Law of Cosines

Activity 13

Activity 13 · from your notes

Now that we can determine which laws to use, solve each triangle (to the nearest degree/whole number).

(a) $B = 112°$, $a = 12$, and $b = 25$.

Answer (a)

$\sin A = \frac{12 \sin 112°}{25} \approx 0.4450 \rightarrow A \approx 26.43°$. (The Quadrant II option won't fit: B is already obtuse.)

$C = 180° - 112° - 26.43° \approx 41.57°$. $c = \frac{25 \sin 41.57°}{\sin 112°} \approx 17.89$

$A \approx 26°$, $C \approx 42°$, $c \approx 18$

(b) $a = 37$, $b = 55$, and $c = 30$.

Hint: Which angle is across from the longest side? Find that one first.

Answer (b)

SSS, so use the Law of Cosines. Start with the largest angle, B (across from 55).

$\cos B = \frac{37^2 + 30^2 - 55^2}{2(37)(30)} = \frac{-756}{2220} \approx -0.3405 \rightarrow B \approx 109.91°$

$\cos A = \frac{55^2 + 30^2 - 37^2}{2(55)(30)} = \frac{2556}{3300} \approx 0.7745 \rightarrow A \approx 39.24°$

$C = 180° - 109.91° - 39.24° \approx 30.85°$

$B \approx 110°$, $A \approx 39°$, $C \approx 31°$

This triangle isn't the one from Activity 12(b). It's a new SSS triangle.

Practice Example 14

Practice Example 14 · from your notes

A triangle has angle $A = 40°$, angle $B = 65°$, and side $a = 12$ (opposite angle A). Find side b (opposite angle B), rounded to two decimal places.

Try it, then check.

Answer

(a, A) is a complete pair, so use the Law of Sines: $\frac{12}{\sin 40°} = \frac{b}{\sin 65°}$

$b = \frac{12 \sin 65°}{\sin 40°} \approx 16.9196\ldots$

$b \approx 16.92$

Practice Example 15

Practice Example 15 · from your notes

A triangle has sides $a = 8$, $b = 10$, and included angle $C = 50°$ (the angle between sides a and b). Find side c, rounded to two decimal places, and explain why the Law of Sines could not be used directly here.

Try it, then check.

Answer

$c^2 = 8^2 + 10^2 - 2(8)(10)\cos 50° \approx 164 - 102.85 = 61.15 \rightarrow c \approx 7.8201\ldots$

$c \approx 7.82$

Why not the Law of Sines? It needs a side and its opposite angle. Here the only known angle, C, is between the known sides, and its opposite side c is the unknown (SAS).

Practice Example 16

Practice Example 16 · from your notes

A triangle has sides $a = 6$, $b = 9$, $c = 11$. Use the Law of Cosines to find angle C (opposite side c), rounded to one decimal place.

Try it, then check.

Answer

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{36 + 81 - 121}{2(6)(9)} = \frac{-4}{108} \approx -0.0370$$

$C = \cos^{-1}(-0.0370) \approx 92.12°$

$C \approx 92.1°$

Summary

FROM THE HANDOUT

Write 2–3 sentences summarizing the key concepts of today's lecture.

Accessibility & help

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